

HW5

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Problem 2

7. Use a direct proof to show that every odd integer is the difference of two squares.

$$n = a^2 - b^2$$

$$n = 2k + 1$$

$$a^2 - b^2 = 2k + 1$$

$$a^2 - b^2 = (a - b)(a + b) = 2k + 1$$

$$a - b = 1$$

$$a + b = 2k + 1$$

$$(a - b)(a + b) = 1 \cdot (2k + 1) = 2k + 1$$

$$\begin{cases} a - b = 1 \\ a + b = 2k + 1 \end{cases}$$

Find a:

$$(a - b) + (a + b) = 1 + (2k + 1)$$

$$a - b + a + b = 2a$$

$$1 + 2k + 1 = 2k + 2$$

$$2a = 2k + 2$$

$$a = \frac{2k + 2}{2} = k + 1$$

$$n = k + 1$$

Find b:

$$a - b = 1$$

$$a = k + 1$$

$$(k + 1) - b = 1$$

$$k - b = 0$$

$$b = k$$

$$a = k + 1, b = k$$

$$a^2 - b^2 = (k + 1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1 = n$$

Problem 3

9. Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.

x = irrational

y = rational

$x + y$ = irrational

proof by contradiction:

$$x + y = r$$

r – is rational

$$x = r - y$$

r -rational, y -rational

$$r = \frac{a}{b} ; y = \frac{c}{d}$$

$$r - y = \frac{ad - bc}{bd} \text{ - rational}$$

$$x = r - y = \text{rational}$$

We have obtained that x is a rational number.

But according to the condition, x is irrational.

Contradiction.

Our assumption (that $x+y$ is rational) is false.

Therefore, $x+y$ is an irrational number.

Problem 4

11. Prove or disprove that the product of two irrational numbers is irrational.

Disproved

$$\sqrt{2} \times \sqrt{2} = 2 \quad \text{rational}$$

$$\sqrt{2} \times \frac{1}{\sqrt{2}} = 1$$

Problem 5

15. Use a proof by contraposition to show that if $x + y \geq 2$, where x and y are real numbers, then $x \geq 1$ or $y \geq 1$.
- $\overbrace{x + y \geq 2}^p$
 $\underbrace{x \geq 1 \text{ or } y \geq 1}_q$

$$p \rightarrow q \quad - \quad \neg q \rightarrow \neg p$$

If $x + y \geq 2 = p$
 then $x \geq 1 \vee y \geq 1 = q$

If $x < 1 \wedge y < 1 = \neg q$
 then $x + y < 2 = \neg p$

Problem 6

17. Show that if n is an integer and $n^3 + 5$ is odd, then n is even using
- a) a proof by contraposition.
 - b) a proof by contradiction.

If $n^3 + 5$ - odd, then n - even

$P: n^3 + 5$ - odd

$Q: n$ - even

$P \rightarrow Q$

a) $\neg Q \rightarrow \neg P$

$\neg Q$: odd

$\neg P$: even

$$n = 2k + 1$$

$$n^3 = (2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1$$

$$n^3 = 2(4k^3 + 6k^2 + 3k) + 1$$

n^3 - odd

$$n^3 + 5 = \text{odd} + \text{odd} = \text{even}$$

$$n^3 + 5 = \text{even} (\neg P)$$

The contraposition has been proven, so the original statement is true.

b)

Proof by contradiction.

Suppose that the statement is false:

$$n^3 - \text{odd}$$

$$n - \text{odd}$$

Since n is odd, as in point (a), n^3 is odd.

$$n^3 + 5 = \text{odd} + \text{odd} = \text{even}$$

We have a contradiction; our proposition is false, so the statement is true.

Problem 8

EXAMPLE 7 Show that if x and y are integers and both xy and $x + y$ are even, then both x and y are even.

$$x, y - \text{integers}$$

$$xy - \text{even}$$

$x+y = \text{even}$
proof x and y are even

Proof by contradiction

Suppose that at least one of the numbers x or y is odd.

1. x – odd, y – even. If x – odd, y – even, then $x + y = \text{odd} + \text{even} = \text{odd}$
2. x – even, y – odd. If x – even, y – odd, then $x + y = \text{even} + \text{odd} = \text{odd}$
3. x – odd, y – odd. If x – odd, y – odd, then $x + y = \text{odd} + \text{odd} = \text{even}$, But $xy = \text{odd} * \text{odd} = \text{odd}$

Conclusion:

All three cases where at least one number is odd led to a contradiction.

Therefore, our assumption is incorrect.

Consequently, both numbers x and y are even.

